(2) Carbon increases the brittleness of the steel, and is mechanically stronger, whereas milder steel is weaker and more ductile. Generally hardness increases and ductility decreases with increased carbon content.

(5) A 5ft section of HP14x102 weighs 510 lbs

(7) Axial load of 200 N, determine stress & deformation of an aluminum wire 20m long & 5mm wide.

$2.5mm^2 \pi = A = 19.63mm^2$

$E = 70GPa \quad \sigma = \frac{200N}{19.63mm^2} = 10.18N/mm^2 = \underline{\underline{1.02\times10^{-5} N/m^2}}$

$\varepsilon = \frac{1.02\times10^{-5} N/m^2}{70GPa} = 1.46\times10^{-13}$

$disp. = 1.46\times10^{-13} \times 20 = \underline{\underline{2.91\times10^{-12}}}$

(11) $\varepsilon = \frac{\sigma}{50}\left[1+\frac{3}{7}\left(\frac{\sigma}{50}\right)^5\right]$

(a) $E = \frac{d\varepsilon}{d\sigma} = \left(\frac{\sigma}{30000} + \frac{\sigma}{70000}\left(\frac{\sigma}{50}\right)^5\right)' = \left[\frac{1}{30000} + \frac{1}{70000}\left(\frac{\sigma}{50}\right)^5 + \frac{\sigma}{70000}\cdot 5\left(\frac{\sigma}{50}\right)^4\right]_{\sigma=0}$

$\frac{d\varepsilon}{d\sigma} = \frac{1}{30000}$

$\frac{\delta l}{.015} = 54000lbs$

$E = \frac{\delta\sigma}{\delta\varepsilon} = 30000ksi$

(b) On spreadsheet

$0.015 = \frac{\sigma}{30000}\left[1+\frac{3}{7}\left(\frac{\sigma}{50}\right)^5\right]$